Design and Evaluation of a Compliant Quasi-Direct Drive End-Effector for Safe Robotic Ultrasound Imaging

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Abstract—**Robot-assisted ultrasound scanning promises to advance autonomous and accessible medical imaging. However, ensuring patient safety and compliant human-robot interaction during probe contact poses a significant challenge. Most existing systems either have high mechanical stiffness or trade performance for compliance. This paper presents a novel compliant end-effector designed to mount on robotic arms for safe and accurate robotic ultrasound imaging, using a quasi-direct drive actuator to achieve passive mechanical compliance and precise active force control. To evaluate the end-effector's performance, we developed an *ex vivo* dynamic motion simulator platform for contact and scanning testing on tissue under simulated movements. The end-effector was evaluated against a UR3e robot arm using conventional force control strategies as a baseline. Single-point contact experiments from 2.5N to 15N show that the end-effector reduced force tracking RMS error by 80.1% on average. Trajectory scanning experiments at different speeds showed an average of 68.0% reduction in force tracking error. Statistically significant improvements were observed in four of six quantitative image quality and stability metrics using the end-effector. The proposed solution provides a novel approach for designing and evaluating compliant robotic ultrasound systems, and is a significant step toward safer, more reliable robotic US systems.**

Index Terms—**Ultrasound, compliance, human-robot interaction, quasi-direct drive, mechatronics design.**

I. Introduction

Ultrasound (US) imaging has been a reliable and versatile clinical tool since the mid-20th century, widely used due to its non-invasive nature, real-time imaging, lack of ionizing radiation, and relatively low cost [1]. However, conventional US exams rely on trained sonographers to manually apply consistent pressure and precisely control the probe's orientation. The increasing demand for US exams has outpaced the growth in the number of sonographers, especially in remote or underserved areas [2] [3]. A 2024 study revealed that between 2011 and 2021, the growth in US examinations (55.1%) outpaced the rise in the number of sonographers (43.6%) [2]. The United States Bureau of Labor Statistics projects a need for an additional 27,600 sonographers by 2024 [4]. In the United Kingdom, 2023 data showed about 416,900 patients were waiting 6 weeks or more for key diagnostic tests that include US, with one-third of patients waiting more than 6 weeks for diagnostic tests require non-obstetric US [5]. Robotic US systems could offer significant advantages by automating probe manipulation and improve imaging precision and reproducibility [6]. Furthermore, robotic US systems can also operate autonomously, enabling US imaging in high-latency or isolated environments such as space missions, remote locations with limited healthcare professionals, or hyperbaric chambers used in naval and commercial diving operations where direct medical supervision may be restricted [6].

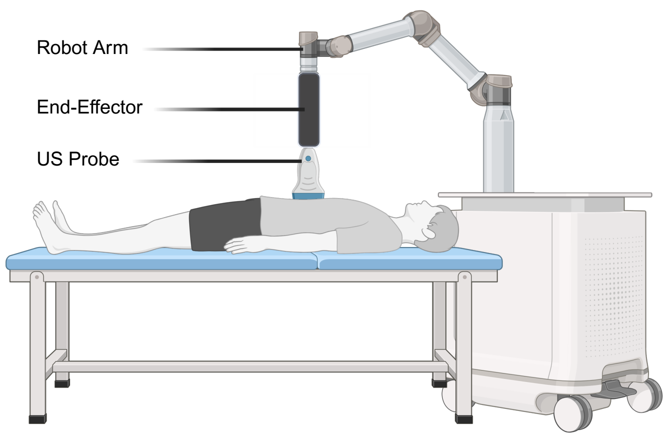

Fig. 1. Schematic diagram of robotic US imaging components, showing a robotic arm with mounted US probe positioned over a patient on an examination table. This direct contact between robotic probe and patient shows the importance of compliance for safe imaging [7].

The safe integration of robotic US systems in clinical practice depends on achieving proper contact compliance: the robot's ability to adapt and yield to patient movements without exerting excessive pressure. A major concern is the potential for excessive force exerted by the robot end-effector, typically rigid and non-mechanically compliant, which can lead to patient discomfort or even injury [8] [9]. This is especially important for abdominal or retroperitoneal imaging, where instinctive movements of the body, such as respiration and tensing in response to touch, would require a robotic system to adapt dynamically to maintain consistent and safe contact with the skin [10]. While patients can temporarily hold their breath, this causes delays in scanning sessions that can last up to 30 minutes. Autonomous systems that can adapt to normal patient movement could shorten scans and make them more comfortable.

Researchers have explored active and passive strategies to address compliance challenges [11]. Active compliance enables a rigid robot arm to exhibit compliant behavior by controlling the end-effector position based on force feedback, often using task-space force controllers to simulate compliant behaviors [12]–[15]. However, these methods are limited by lower bandwidths, making them slower to respond to sudden movements [16]. In addition, most commercial rigid robot arms only support position or velocity control modes. This

TABLE I
COMPARATIVE OVERVIEW OF STATE-OF-THE-ART COMPLIANT ROBOTIC US SYSTEMS AND OUR PROPOSED SYSTEM

	Operating Force Range[e]	Operating Position Range	Feedback Types	Active Force Regulation	Mechanical Compliance
Our Proposed System	**2.5N - 15N**	**52mm**	**Position, Force**	**Yes**	**Yes**
Mathiassen16[a] [12]	1N - 5N	N/A	Position, Force	Yes	No
Lindenroth17[b] [17]	0N - 10N	6.3mm	None	No	Yes
Wang23[a] [14]	3N - 15N	N/A	Position, Force	Yes	No
Bao21[c] [18]	4N - 24N	8mm	Position, Force	Yes	Yes
Wu25[c] [19]	12.3N - 21.3N	3mm	Position, Force	Yes	Yes
Kuo23[d] [8]	4.1N - 7.5N	50mm	Force	Yes	Yes

[a]Robot arm only systems; [b]Soft actuator; [c]Elastic actuator; [d]Pneumatic actuator.
[e]Clinical requirements for force: 7.5N mean, 17.3N max [22]
Pain/discomfort threshold lower-bound: 20.0N (extrapolated) [23]

restriction forces compliance to be implemented at a higher control level through position or velocity corrections, resulting in slower force response times and reduced stability compared to systems with native current or torque control capabilities. Passive compliance, achieved by integrating mechanically compliant elements, allows the robot to physically deflect or deform in response to external forces without a controller. This includes soft materials, elastic actuators, and pneumatic actuators [8], [17]–[19]. However, each approach has significant limitations. Soft end-effectors can apply variable force based on deformation, while linear elastic actuators have very limited operational position range due to the physical constraints of spring deflection limits [18] [19]. Pneumatic actuators can regulate force but are nonlinear, with limited control bandwidth [20] [21]. Additionally, some approaches [17] [8] lack position measurement capability, which is needed for 3D reconstruction of robotic US scans.

While electric motors offer improved bandwidth and ease of control compared to pneumatic and fluidic actuators, traditional electric motors used in robotic arms often lack mechanical compliance due to their high gear ratios and significant backdrive force [24]. However, recent advancements have led to the development of quasi-direct drive (QDD) actuators, which can achieve high torque control fidelity while maintaining low backdrive torque due to their low gear ratios. This design allows external forces to backdrive the motor and achieve compliance mechanically [25].

For evaluation, a natural human movement mimicking platform is needed to perform robust and realistic evaluation of compliant US systems. While several approaches have been developed to ensure safe and effective force control during US procedures [8], [12]–[15], [17]–[19], existing testing tissue platforms are predominantly static. As a result, current tissue platforms have limited capability to accurately replicate the dynamic behavior of both physiological and unexpected human motion during abdominal or retroperitoneal US exams. This gap in available testing infrastructure restricts the ability to fully evaluate the performance of compliant US systems in realistic scenarios.

This paper proposes a novel single-degree-of-freedom end-effector attachment that mounts to robotic arms for safe US imaging. Our design uses QDD actuation to achieve several key advantages: passive mechanical compliance ensures safe patient interaction even in the event of control failure; precise active force control limits excessive force and enables fast real-time adjustments to dynamic movements; and spatial feedback of the probe's position is maintained across a large operating range. To evaluate this end-effector, we propose an actuated *ex vivo* tissue platform to simulate the motion of the human abdomen or back during breathing and sudden unexpected movements. Unlike standard static phantom testing, this dynamic platform provides a more realistic evaluation environment and could be a general testing methodology for evaluating other robotic US systems beyond our specific end-effector design.

II. METHODS

A. Clinically Informed Design Requirements

Clinical US imaging requirements inform our design specifications, focusing on key metrics often inadequately addressed by existing solutions: operating position and force range, active force control, probe position feedback, and mechanical compliance. Force capability presents a particular challenge. For normal weight patients, sonographers typically apply contact forces of 7.5 N (mean) and 17.3 N (maximum) during US scans [22]. However, patient comfort limits must also be considered. Waller et al. [23] reported pressure pain threshold reference values in pain-free older adults, with the lowest recorded threshold of 75 kPa. When extrapolated to the US transducer's contact surface used in this study (Interson SP-L01, Interson Corporation, Pleasanton, CA) of 38mm by 7mm, this threshold corresponds to approximately 20.0 N of applied force. As shown in Table I, some state-of-the-art compliant US end-effectors fall short of reaching the clinically required force levels while remaining within patient comfort limits.

B. High-Performance Compliant Actuation

Achieving both passive compliance and precise active force control is a fundamental challenge in robotic actuation. While various actuation approaches have been used in state-of-the-art solutions, each comes with trade-offs. This section compares these established actuation methods and shows why we selected QDD actuators, which address the limitations of existing approaches while maintaining the dual requirements of compliance and control fidelity.

Compared to pneumatic actuators, QDD systems eliminate complex pressure regulation systems, which are often nonlinear and exhibit slower response times [20] [21]. Additionally, QDD actuators provide built-in positional feedback [26], eliminating the need for external position sensors.

Compared to elastic actuators, QDD actuators avoid the fundamental trade-off between control fidelity and compliance. While elastic actuators must balance these characteristics through spring elements, QDD actuators achieve both high control fidelity and backdrivability simultaneously through their low gear ratio design [27]. This approach also reduces system complexity by eliminating springs and external torque sensors, as accurate torque estimation can be performed using motor current [28].

Compared to soft fluidic actuators, QDD systems provide more predictable behavior and simplified control strategies, avoiding the unpredictable deformation and complex feedback systems required for maintaining force accuracy in soft systems [29].

QDD actuators have been successfully implemented in various fields requiring compliance, safe human-robot-interaction, and impact absorption, including powered prostheses [30], powered orthoses [31], and legged robotics [16] [26]. For this study, we chose the AK60-6 (Cubemars, Shenzhen, China) QDD actuator, with a rated torque of 3 Nm, a backdrive torque of 0.2Nm, and built-in positional feedback.

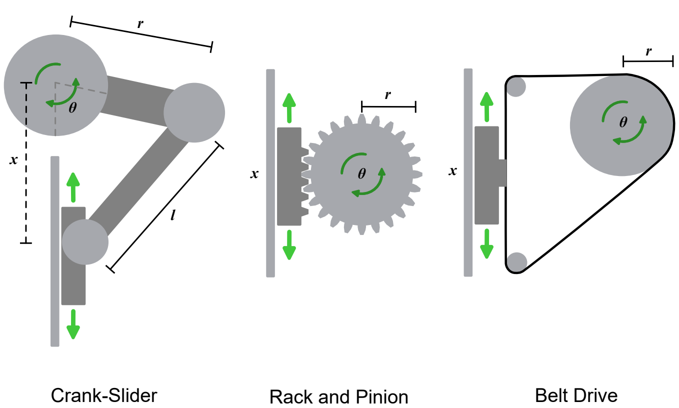

Fig. 2. Comparative analysis of three mechanical transmissions for rotational-to-linear motion conversion. From left to right: crank-slider mechanism, rack and pinion gear system, and belt drive system. Each mechanism is illustrated with its kinematic configuration (green arrows indicate direction of motion) and operational parameters, the section shows the belt drive's advantages of linear motion transmission, lower measurement sensitivity, and inherent compliance for robotic US applications.

C. Transmission Selection for Optimal Control and Safety

Converting the QDD's rotational motion into linear motion requires a transmission that preserves the actuator's strengths in backdrivability and control precision, while complementing the safety/compliance capabilities. This section describes the primary design considerations and compares multiple transmission types we evaluated. We eliminated lead screws (poor backdrivability) and ball screws (excessive cost), then evaluated three viable candidates: crank-slider, rack and pinion, and timing belt driven systems, as shown in Fig. 2. The belt-driven transmission ultimately met our requirements best, maintaining backdrivability while eliminating the backlash and kinematic limitations present in the other mechanisms.

1) Crank-Slider: Our first prototype used the crank-slider mechanism, but we ultimately transitioned away from it due to its limitations. While capable of high force transmission, the crank-slider exhibits nonlinear motion characteristics with variations throughout its stroke. By the principle of virtual work, force transmission is the inverse of displacement transmission, so this nonlinearity can make controllers more challenging to design, and can be understood through its kinematic equation

$$x = r\cos\theta + \sqrt{l^2 - r^2\sin^2\theta} \tag{1}$$

where x is the linear displacement, r is the length of the crank, l is the length of the connecting rod, and θ is the angle of rotation. The derivative of this relationship

$$\frac{dx}{d\theta} = -r\sin\theta - \frac{r^2\sin\theta\cos\theta}{\sqrt{l^2 - r^2\sin^2\theta}} \quad \text{(variable)} \tag{2}$$

reveals that the rate of change of the linear displacement varies nonlinearly with θ, thus creating control challenges, especially at the extremes of motion where mechanical advantage approaches zero.

A sensitivity analysis reveals three measurements contributing to uncertainty: σ_r, σ_l, and σ_θ. The partial derivatives with respect to system parameters are

$$\frac{\partial x}{\partial r} = \cos\theta - \frac{r\sin^2\theta}{\sqrt{l^2 - r^2\sin^2\theta}} \tag{3}$$

$$\frac{\partial x}{\partial l} = \frac{l}{\sqrt{l^2 - r^2\sin^2\theta}}. \tag{4}$$

The third term $\frac{\partial x}{\partial \theta}$ is derived in Eq. 2.

The total position uncertainty results from the combination of these error sources according to

$$\sigma_x^2 = \left(\frac{\partial x}{\partial r}\right)^2 \sigma_r^2 + \left(\frac{\partial x}{\partial l}\right)^2 \sigma_l^2 + \left(\frac{\partial x}{\partial \theta}\right)^2 \sigma_\theta^2. \tag{5}$$

This error propagation exhibits nonlinear position dependence, meaning measurement errors will produce varying positional uncertainties depending on crank angle.

Furthermore, the rigid nature of the system lacks inherent mechanical compliance, reducing the tolerance for safe human-robot-interaction.

2) Rack and Pinion: The rack and pinion system offers excellent positioning accuracy and a direct linear relationship between input rotation and output motion. However, this rigid mechanical coupling transmits all impact forces directly to the actuator and like the crank-slider mechanism, also lacks inherent mechanical compliance. In addition, the linear input-output characteristics are not unique to this transmission, as belt-driven systems also exhibit this behavior.

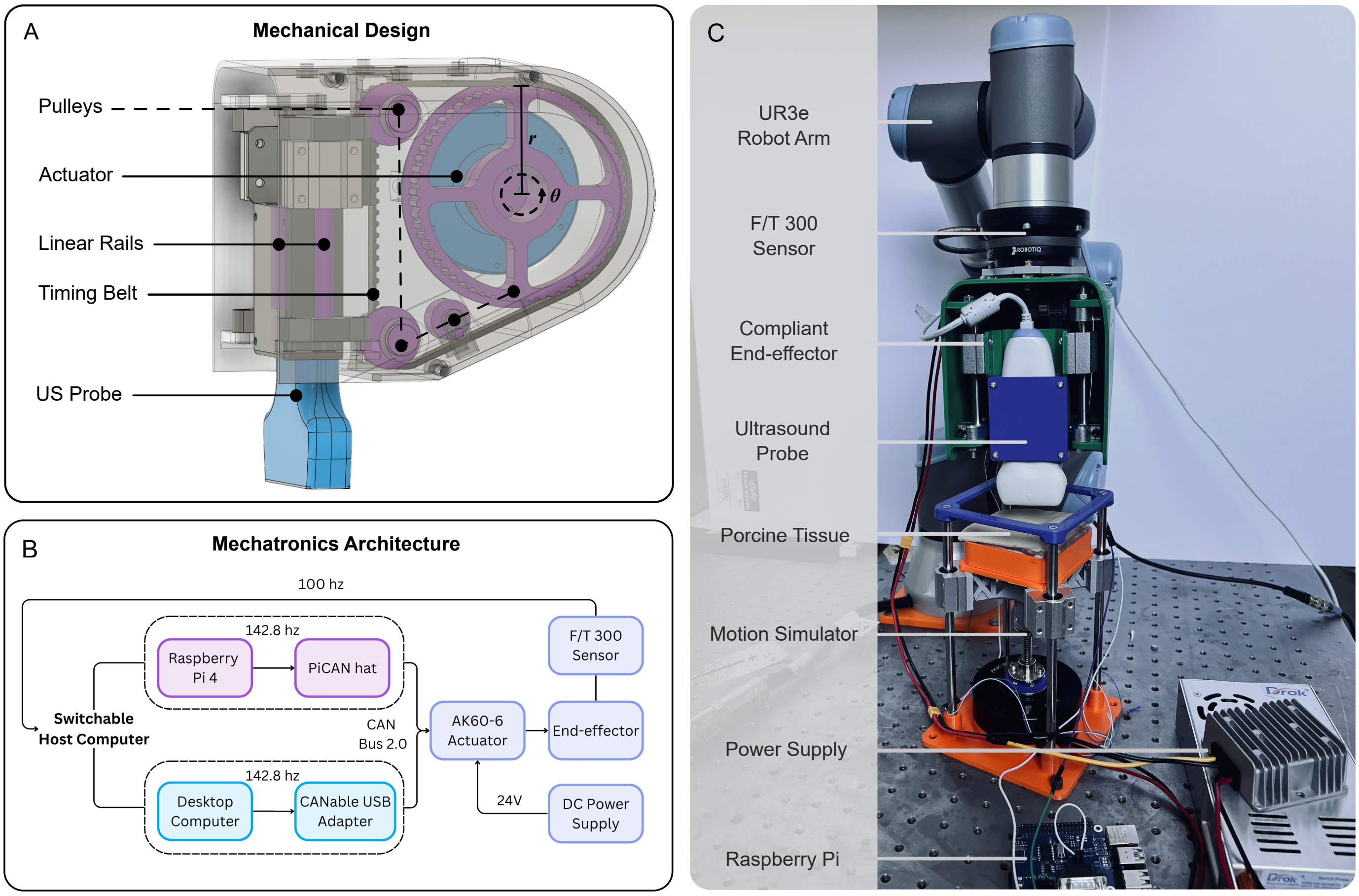

Fig. 3. Overview of the mechatronic system design and setup, demonstrating the complete development pipeline from mechanical design through electronics architecture to experimental validation. A: schematic diagram breakdown of the compliant robotic US end-effector. The QDD actuator drives a large-diameter pulley with radius r. The rotational angle θ translates to linear motion via a timing belt driving the US probe mount on linear rails, a tensioner pulley is used to maintain belt tension; B: mechatronics architecture of the end-effector. The mechatronics setup also powers the dynamic motion simulator; C: experiment setup of the end-effector attached to the UR3e robot arm in the point contact setup, enabling evaluation of the system under realistic conditions

3) Belt-Driven Transmission: In contrast, the belt-driven transmission system provides several distinct advantages. Like the rack and pinion system, the timing belt system has a simpler force transmission compared to the crank-slider linkages, resulting in a more linear relationship, as described by:

$$x = r\theta \tag{6}$$

where x is the linear displacement, r is the radius of the pulley, and θ is the angle of rotation. The derivative,

$$\frac{dx}{d\theta} = r \quad \text{(constant).} \tag{7}$$

shows a constant rate of change, resulting in a direct linear relationship between input rotation and output motion.

The sensitivity analysis for the belt system yields,

$$\frac{\partial x}{\partial r} = \theta \tag{8}$$

$$\frac{\partial x}{\partial \theta} = r \tag{9}$$

with error propagation,

$$\sigma_x^2 = \theta^2\sigma_r^2 + r^2\sigma_\theta^2. \tag{10}$$

The relative error $\frac{\sigma_x}{x} = \frac{\sigma_r}{r} + \frac{\sigma_\theta}{\theta}$ shows that errors combine linearly without position-dependent amplification. Error propagation is contributed by only two terms, r and θ, compared to three in the crank-slider linkage. The additional linkage length l in the crank-slider introduces another measurement error source, increasing the overall error propagation. The derivative in Eq. 7 confirms the system's linearity, which provides predictable control characteristics throughout the entire range of motion, in contrast to the crank-slider's nonlinear behavior. This difference in sensitivity characteristics makes the belt system more suitable for applications requiring consistent positioning accuracy and straightforward control implementation.

Beyond this, the inherent properties of the belt material provide intrinsic impact absorption, adding compliance to the system over the other methods. The belt system also requires less precise manufacturing tolerances than the other two systems and provides natural overload protection through controlled slippage.

While the above calculations assume a rigid belt without stretch, real belt systems exhibit a very slight elongation under load and require periodic tension adjustment. Belt stretch will be incorporated later when formulating the dynamic equations of the system. However, the nonlinearity introduced by belt stretch is minimal compared to the crank slider mechanism, and the superior compliance and simplified control characteristics of the belt drive make it well-suited for this application.

D. Design

The complete design is illustrated in Fig. 3A. The actuator drives a 96.5mm diameter pulley, which is coupled to a linear slide mechanism housing the US probe. The probe has a maximum travel distance of 52mm, sufficient to accommodate abdominal movement even during deep breathing [32]. The housing is fabricated from PLA plastic. With a total weight of 1.13 kg, the system is compatible even with small robot arms such as the UR3e, which has a maximum payload of 3 kg. When fully retracted, the system measures only 178mm in length (including the US probe), making navigation around the human body easier compared to longer end-effectors. The backdrive force, which represents the minimum external force required to move the probe when the system is not powered, is calculated to be 4.1N.

E. Dynamic Model of the QDD End-Effector System

To understand how forces and motions propagate through the system, we need a mathematical model of the complete actuator-transmission-end-effector chain. This section derives a model to analyze the dynamics and predict system behavior.

The end-effector is governed by the coupled dynamics of the QDD actuator and timing belt transmission. The motor dynamics follow the fundamental torque balance equation,

$$J_m\ddot{\theta} + b_m\dot{\theta} = K_t I - \frac{\tau_l}{N} \tag{11}$$

obtained from the manufacturer's datasheet, where $J_m = 2.435 \times 10^{-5}$ kg·m^2 is the motor inertia, $b_m = J_m/\tau_{\text{mech}} = 0.03$ Nm·s/rad is the viscous damping coefficient derived from the mechanical time constant $\tau_{\text{mech}} = 0.81$ ms, $K_t = 0.078$ Nm/A is the torque constant, $N = 6$ is the gear ratio, I is the motor current, θ is the motor shaft angle, and τ_l is the load torque at the output shaft. The timing belt introduces compliance between the motor output and the linear probe motion,

$$F_{\text{belt}} = k_b\left(\frac{r\theta}{N} - x\right) \tag{12}$$

where $k_b = 2013.8$ N/m is the nominal stiffness of a GT2 timing belt under moderate (low-strain) tension [33], $r = 0.04825$ m is the pulley radius, and x is the probe's true linear displacement. Applying Newton's second law to the probe assembly yields

$$m_{\text{eff}}\ddot{x} = F_{\text{belt}} - F_{\text{contact}} + F_g \tag{13}$$

where $m_{\text{eff}} = m_{\text{probe}} + \frac{J_m N^2}{r^2}$ is the effective mass including the reflected motor inertia, F_{contact} is the tissue interaction force measured by the force sensor, and $F_g = m_{\text{probe}} g \cos\alpha$ is the gravitational force component along the probe axis, with α being the angle between the probe axis and the vertical direction.

Combining these equations yields the complete system dynamics. Defining the state vector $\mathbf{x} = [\theta, \dot{\theta}, x, \dot{x}]^T$ and inputs $\mathbf{u} = [I, F_{\text{contact}}]^T$ (where I is the control current input and F_{contact} is the measured contact force), the system can be written in linear state-space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{c} \tag{14}$$

where:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{r^2 k_b}{J_m N^2} & -\frac{b_m}{J_m} & \frac{r k_b}{J_m N} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{r k_b}{m_{\text{eff}} N} & 0 & -\frac{k_b}{m_{\text{eff}}} & 0 \end{bmatrix} \tag{15}$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 \\ \frac{K_t}{J_m} & 0 \\ 0 & 0 \\ 0 & -\frac{1}{m_{\text{eff}}} \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{F_g}{m_{\text{eff}}} \end{bmatrix}. \tag{16}$$

This linear fourth-order model describes the dynamic model of the compliant end-effector.

F. Mechatronics Architecture

The mechatronics system architecture (Fig. 3) supports two host computer configurations: (1) a stationary desktop computer with a CANable USB-CAN adapter (Protofusion, Ellicott City, MD), or (2) a mobile Raspberry Pi 4 with a PiCAN2 hat (Copperhill Technologies, Greenfield, MA). Both configurations use CAN bus 2.0 for communication at speeds up to 1 Mbps. A 48V power supply feeds a buck converter, which steps down the voltage to 24V to drive the AK60-6 QDD actuator. Force measurements are obtained using a six-axis F/T 300 force sensor (Robotiq, Lévis, QC, Canada), sampling at 100 Hz. The control loop executes at 142.8 Hz. While current implementations use stationary power sources, the Raspberry Pi-based system is designed to be easily adapted for mobile operation by incorporating a battery pack, making this system suitable for remote or resource-constrained environments. The system architecture also powers our dynamic motion simulator. Experiments were conducted on both the desktop computer and Raspberry Pi configurations.

G. Controller Design

To evaluate the end-effector's performance compared to a robot arm only baseline, we implemented simple controllers to isolate mechanical benefits while accounting for system-specific control input limitations (position, velocity, current). This section describes the three controllers used in our experiments.

As illustrated in Fig. 4, Controller A is implemented on the end-effector, while Controllers B and C are implemented on the robot arm. Controller A and B are Proportional-Integral-Derivative (PID) controllers, a widely adopted approach that does not necessitate prior model knowledge [34]. Controller C is an admittance controller.

The controller in Fig. 4A is a PID current controller implemented for the end-effector system. We chose to implement a simple PID controller instead of a model-informed controller using the dynamic model derived in Eq. 14-16 to better isolate the effects of mechanical design from control implementation.

Direct current control is made possible by the mechanical design of the QDD actuator, which uses a low gear ratio that enables direct current-based force estimation and control. The low gear ratio minimizes mechanical impedance and allows for accurate force estimation through motor current measurements [35].

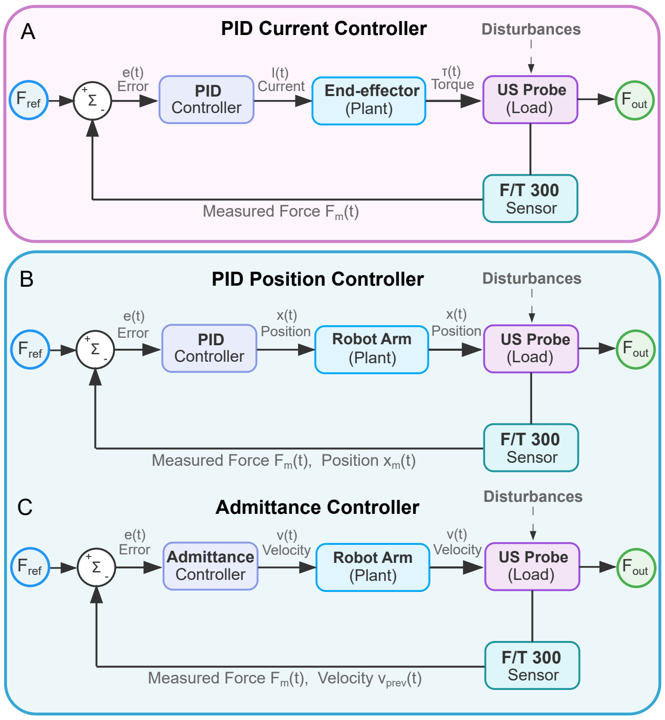

Fig. 4. Controller architecture for both the end-effector and the robot arm. Only one controller is active during their respective experiments. A: the end-effector's current-based PID force controller; B: the robot arm's position-based PID force controller; C: the robot arm's admittance controller. These different control strategies provide comprehensive assessment of multiple methods to achieving compliant behavior.

The controllers in Fig. 4B-C were implemented only on the robot arm platform to serve as comparative baselines during experimental trials. During each experimental trial, only one of these controllers was active to ensure clear performance evaluation. The controller in Fig. 4B uses a PID position-based force control strategy, where force regulation is achieved through adjusting the positions sent to the robot arm's joints. The controller uses position-based commands as commercial robot arms typically do not allow direct current control. This approach converts force errors into position corrections. The controller implementation was based on the controller from [15]. To optimize both PID controllers, we first used the Ziegler-Nichols tuning method as a starting point, and then tuned heuristically based on feedback. The gains were tuned until the best performance was observed for each condition. The gains used in the experiments are shown in Table II.

The controller in Fig. 4C implements an admittance control architecture, which computes the desired velocities based on the measured force errors. This approach allows the robot to exhibit compliant behavior by effectively reducing its apparent stiffness in response to external forces. Unlike the position-based approach, admittance controllers directly control the velocity. The controller implementation was based on the controller from [14].

TABLE II
PID GAIN VALUES

	k_p	k_i	k_d	**Force (N)**	**Experiment**
1	0.629	8.85	0.015	2.5	**End-effector**
2	0.529	12.85	0.015	5	
3	0.429	8.85	0.015	10	
4	0.429	8.85	0.015	15	
5	0.03	5×10^{-8}	0.001	2.5	**Robot arm only**
6	0.03	5×10^{-8}	0.001	5	
7	0.03	5×10^{-8}	0.001	10	
8	0.025	5×10^{-8}	0.001	15	
9[a]	0.26	2.39	0.01	5	**Sudden movement**
10[b]	0.02	5×10^{-8}	0.001	5	

[a]End-effector; [b]robot arm only

For the trajectory scan experiments, we implemented a path planning algorithm with a multi-segment trajectory generation scheme where the robot arm's tool link follows a sequence of waypoints $\mathbf{p}_i$ with corresponding surface normals $\mathbf{n}_i$. For each segment $i \to i+1$, the translational speed is computed as

$$\mathbf{v}_{\text{trans}} = \frac{\mathbf{d}}{t_{\text{segment}}} \tag{17}$$

where $\mathbf{d} = (\mathbf{p}_{i+1} + \mathbf{n}_{i+1}\ell) - (\mathbf{p}_i + \mathbf{n}_i\ell)$ represents the direction vector, ℓ is the probe length, and $t_{\text{segment}} = \frac{\|\mathbf{d}\|}{v_{\text{path}}}$ is the segment duration based on the prescribed path speed v_{path}. The rotational motion is parameterized using axis-angle representation, where the rotation axis is

$$\mathbf{k} = \frac{(-\mathbf{n}_i) \times (-\mathbf{n}_{i+1})}{\|(-\mathbf{n}_i) \times (-\mathbf{n}_{i+1})\|} \tag{18}$$

and the rotation angle is

$$\theta = \arctan 2(\|(-\mathbf{n}_i) \times (-\mathbf{n}_{i+1})\|, (-\mathbf{n}_i) \cdot (-\mathbf{n}_{i+1})) \tag{19}$$

yielding an angular velocity $\boldsymbol{\omega} = \frac{\mathbf{k}\theta}{t_{\text{segment}}}$.

During baseline trials without the end-effector, the algorithm superimposes the admittance controller in Fig. 4C as the normal velocity with the translational velocity

$$\begin{aligned} \mathbf{v}_{\text{total}} &= \mathbf{v}_{\text{trans}} + \mathbf{v}_{\text{norm}} \\ \mathbf{v}_{\text{norm}} &= v_{\text{norm}}\mathbf{n}_{\text{current}} \\ v_{\text{norm}} &= c_1(F_m - F_{\text{target}}) + c_2 v_{\text{prev}} \end{aligned} \tag{20}$$

to maintain constant contact force, where c_1 and c_2 are controller gains, F_m is the measured normal force, F_{target} is the desired contact force magnitude, and v_{norm} clamped to $\pm v_{\max}$. During trials with the end-effector, the admittance controller is turned off.

H. *Ex Vivo Dynamic Motion Simulator Design*

This section describes the development of an actuated *ex vivo* tissue platform that addresses a key limitation in robotic ultrasound testing: static phantoms cannot replicate the tissue motion from patient movements. We designed the platform

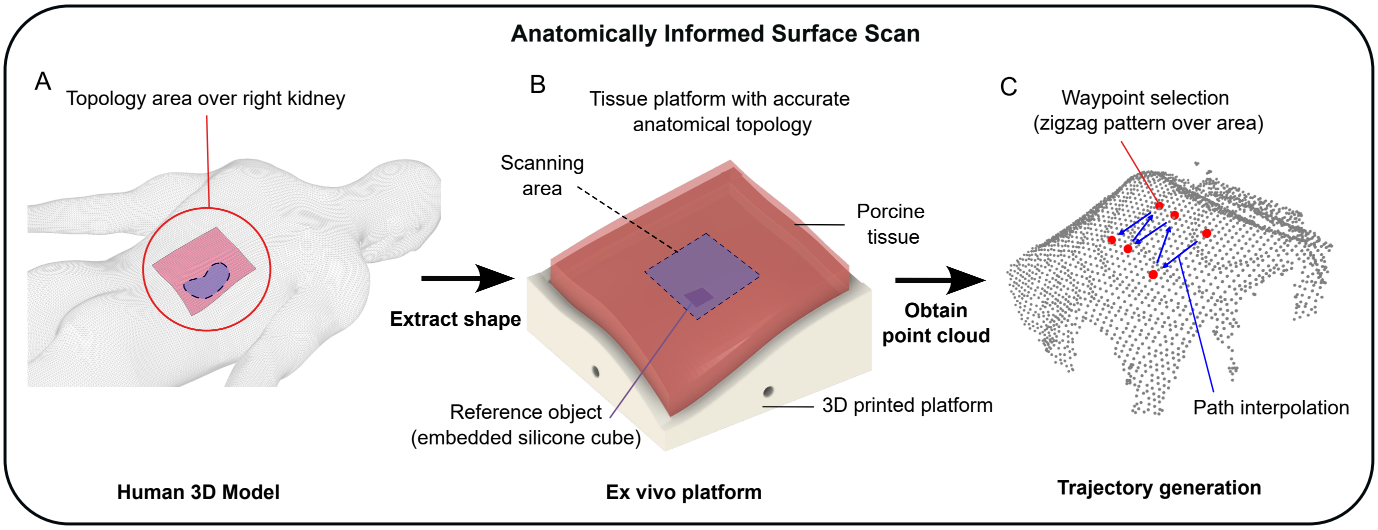

Fig. 5. Anatomically informed testing platform development. A: human male 3D model with a highlighted posterior section above the right kidney; B: 3D-printed platform replicating uneven/non-flat topography of a human back, featuring a central silicone-filled cavity as US reference target, with porcine abdominal tissue on top; C: point cloud surface reconstruction obtained via depth camera scanning, with overlaid zigzag trajectory path for US probe navigation. This platform provides comprehensive evaluation by combining anatomically realistic, non-flat surface geometry with active motion during scanning, providing an evaluation environment under realistic clinical conditions

to mimic the dynamic movements of a human abdomen (while supine) or retroperitoneal region (while prone) during respiration to achieve realistic validation.

The simulator has two broad components: a tissue staging platform, and an actuation system to simulate the body's movement during respiration. The tissue platform is designed to hold both *ex vivo* tissue and tissue phantoms. An AK80-9 actuator drives a lead screw, actuating the tissue platform to move in an axial direction, following a similar lead screw mechanism as the the system in [36]. The simulator is shown in Fig. 3C. During testing, the platform oscillates in a sinusoidal pattern with an amplitude of 9.9 mm and a frequency of 14.6 cycles per minute, simulating human abdominal movements characteristic of quiet breathing in a supine position [32].

To perform realistic experiments on the proposed end-effector, two types of materials were tested: a tissue phantom and *ex vivo* porcine abdominal wall tissue purchased from a local supermarket. The porcine tissue comprised skin, subcutaneous fat, and underlying muscle layers. This allowed us to test the end-effector on both homogeneous tissue phantoms and heterogeneous tissue specimens similar to human abdomen tissue. For tissue phantom creation, 2.5% high-acyl gellan gum (% mass) (VWR, PA, United States) was used following the procedure previously proposed [37]. The developed phantom has Young's modulus greater than 25 kPa, extrapolating values presented in [37].

The point-contact experiments, which simulated abdominal ultrasound exams, used a flat surface to mount the tissue. In contrast, the trajectory-scan experiments were designed to emulate renal examinations, which not only involve scanning over a curved, uneven anatomical surface but also must contend with the rise and fall of the skin due to respiratory motion. We selected the renal exam scenario because of its prevalence in clinical practice and the additional compliance challenges it presents. A three-dimensional human male 3D model, reconstructed from medical imaging data [38], provided the basis for creating an anatomically accurate testing platform. A square section positioned directly above the right kidney on the posterior surface was extracted from this model and 3D printed using PETG plastic to replicate the natural topographical features of the human back. The printed platform incorporated a cubic cavity at its center, which was filled with Dragonskin medium silicone (Smooth-On Inc, PA, United States) to serve as both a reference object and target of interest during US imaging. Porcine abdominal wall tissue, cut to match the platform dimensions, was mounted over the surface with US coupling gel applied at the tissue-platform interface to ensure acoustic continuity. The tissue was then covered with fabric and scanned using an Intel RealSense SR305 depth camera (Intel, Santa Clara, CA, United States) to generate a three-dimensional point cloud representation of the surface topography. Finally, we selected six points in a zigzag pattern to cover an area roughly the size of an adult human kidney.

III. Experiments and Results

A series of force tracking experiments were done to evaluate the performance of our compliant end-effector while mounted on a UR3e robot arm (Universal Robots, Odense, Denmark) compared to the UR3e robot arm alone without the end-effector. The experimental setup is shown in Fig. 3C.

Two sets of experiments were conducted. The first set consisted of point contact experiments that simulates abdominal exams, where the contact point remains fixed and the goal is to maintain steady force while subjected to simulated respiratory motion (9.9mm displacement, 14.6 breaths per minute [32]) on the dynamic motion simulator. We used the robot arm alone as the baseline condition, seen in Fig. 7. During the

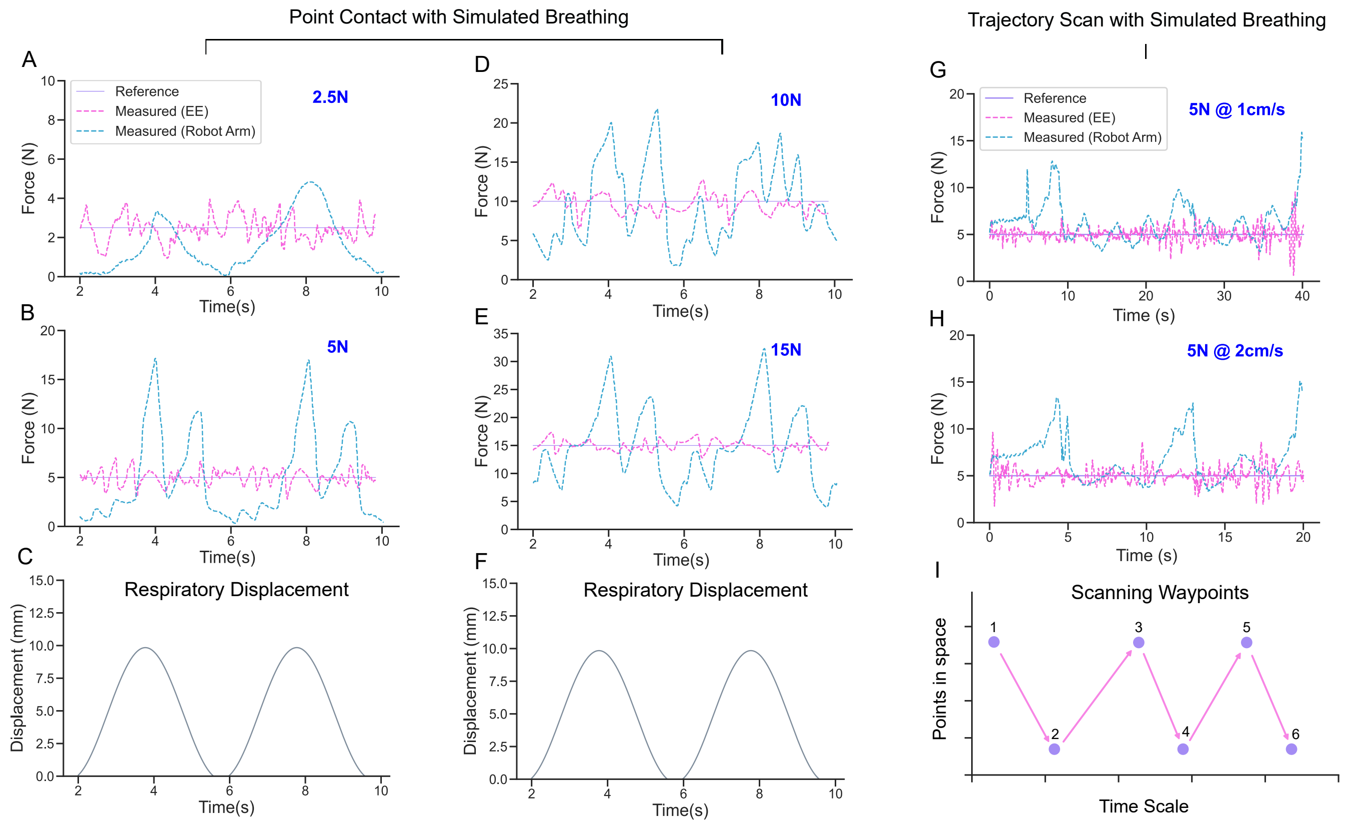

Fig. 6. Experiment setup and force tracking results on porcine tissue with dynamic movement. A, B, D, E: force tracking plots of the end-effector (EE) and robot arm only conditions targeting 2.5N, 5N, 10N, and 15N during simulated breathing motion on point contact; C, F: the displacement caused by the dynamic motion simulator during the experiment; G, H: force tracking plots of the end-effector and robot arm only conditions targeting 5N during simulated breathing motion on trajectory scan at 1cm/s and 2cm/s; I: indicates where waypoints occur along the scanning trajectories. The end-effector consistently shows superior force tracking performance with reduced overshoot and better target maintenance compared to the robot arm only configuration across all tested conditions.

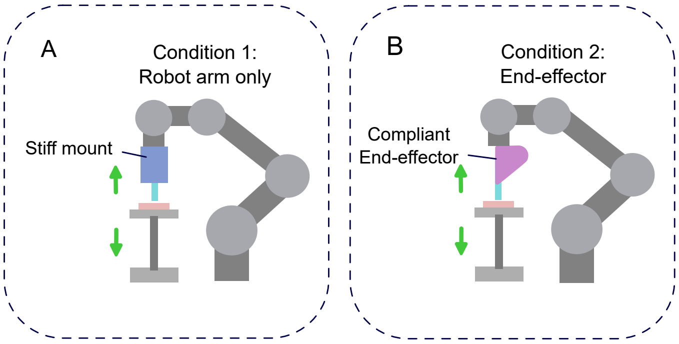

Fig. 7. Experimental setup. A, Robot arm only baseline condition with rigid end-effector; B, compliant end-effector condition with the end-effector attached to the robot arm. Both setups replicate realistic ultrasound scanning conditions for comprehensive system evaluation.

robot arm only conditions, the compliant end-effector was replaced with a rigid one, and the robot arm used the position-based PID controller from Fig. 4B. During the end-effector conditions, the robot arm was not powered, and the end-effector used the current-based PID controller from Fig. 4A. The tuned PID values used for each experiment can be found in Table II. Force measurements were obtained using a six-axis F/T 300 force sensor. Data collection involved recording force measurements at target forces of 2.5N, 5N, 10N, and 15N over 8-second intervals (two complete respiratory cycles), with n=3 measurements obtained for each condition. Mean values and root mean square error (RMSE) were calculated to assess force tracking accuracy. We defined forces above 150% of the target force as “excessive force”, measuring this excessive force duration as a percentage of total operation time.

The second set of experiments added trajectory scanning in addition to the simulated respiratory motion on the dynamic motion simulator. Instead of maintaining a single contact point, the anatomically informed testing platform from the previous section was used as the surface to perform a trajectory scan. The robot arm followed a zigzag path covering a square area approximately the size of a male kidney. During the robot arm only conditions, the arm is fitted with a rigid end-effector, and used the admittance controller from Fig. 4C. During the end-effector conditions, the robot arm followed the scanning path without feedback-related adjustments, and the end-effector independently adjusted the contact force using the controller from Fig. 4A. All experiments were performed at 5N, with path speeds of 1cm/s and 2cm/s compared to assess their effect on force tracking accuracy. Each condition was repeated n=3 times, with mean values and RMSE calculated to assess force

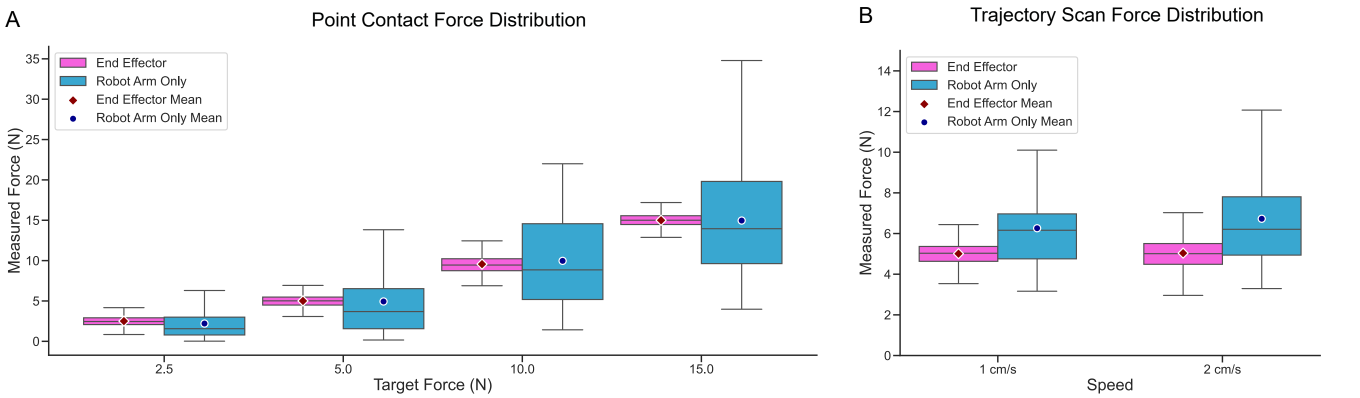

Fig. 8. Force distribution box plots showing concatenated data from three measurements at each condition; A: point contact experiments at 2.5N, 5N, 10N, and 15N, with compliant end-effector and robot arm only conditions compared side by side; B: trajectory scan experiments at 1cm/s and 2cm/s, with compliant end-effector and robot arm only conditions compared side by side. The compliant end-effector consistently exhibits tighter force distributions with reduced variability and better centering around target values compared to the robot arm only configuration.

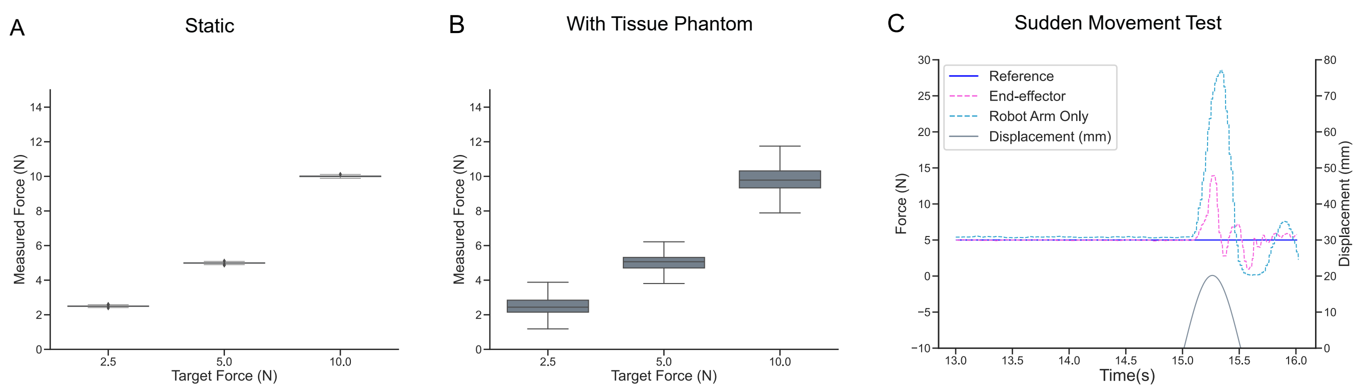

Fig. 9. Force distribution boxplots and force tracking plot of supplementary experiments; A: force distribution of static point contact at 2.5N, 5N, and 10N using the compliant end-effector; B: Force distribution of dynamic point contact at 2.5N, 5N, and 10N on a tissue phantom using the compliant end-effector; C: force tracking comparison plots between compliant end-effector and robot arm only conditions under sudden movement while targeting 5N. These supplementary experiments show the end-effector's consistent performance across exploratory conditions, with clear superiority in handling sudden disturbances compared to the robot arm only configuration.

tracking accuracy.

Three supplementary experiments were conducted as exploratory scenarios, distinct from the primary experimental protocol: (1) assessment of the compliant end-effector's positional accuracy on porcine tissue under static conditions (i.e., with dynamic motion disabled), (2) evaluation of the compliant end-effector's performance during dynamic motion using a tissue phantom as a substitute for biological tissue, and (3) comparative analysis of the compliant end-effector and the robotic arm alone in response to a sudden jerking movement induced by the tissue.

A. Point Contact Experiments

1) Force Tracking on UR3e Robot Arm with PID controller: The UR3e robot arm without the compliant end-effector served as the baseline condition, tested at force levels of 2.5N, 5N, 10N, and 15N while subjected to simulated respiratory motion. The measured mean forces, RMSE, and percentage of time above 150% of the target force for each target force can be found in Table III. The large RMSE and high forces reflect the challenges of maintaining precise force control using conventional force control on the robot arm alone, especially in dynamic scenarios. See Fig. 6(A, B, D, E) for the force tracking plots in a singular measurement, and Fig. 8A for the distribution of all measurements.

TABLE III
FORCE TRACKING PERFORMANCE COMPARISON UNDER SIMULATED RESPIRATORY MOTION

Force	**2.5 N**	**5.0 N**	**10.0 N**	**15.0 N**
Robot Arm Only				
Mean (N)	2.20	4.94	9.97	14.96
RMSE (N)	2.09	4.30	5.43	6.99
% time above 150% of target	15.29	21.33	23.21	16.95
w/ Compliant End-effector				
Mean (N)	2.50	5.01	9.56	14.99
RMSE (N)	0.61	0.77	1.09	0.86
% time above 150% of target	2.65	0.00	0.00	0.00
RMSE reduction (%)	70.8	82.1	79.9	87.7

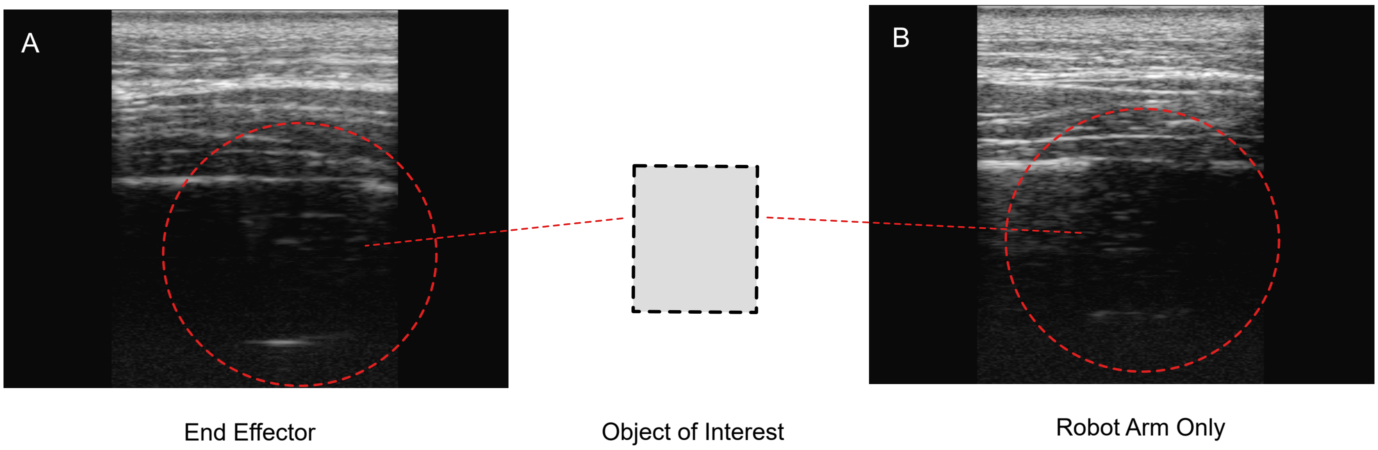

Fig. 10. Representative US frames captured during trajectory scanning experiments showing the reference object (an embedded silicone cube beneath porcine tissue) for A: the compliant end-effector condition and B: the robot arm only condition. Both approaches successfully visualized the target silicone cube.

2) Force Tracking on Compliant End-effector: The end-effector was tested under the same simulated breathing conditions and target forces. The measured mean forces, RMSE, and percentage of time above 150% of the target force for each target force can be found in Table III. This shows an average RMSE reduction of 80.1% compared to the robot arm only condition across all four force levels. These results showed the system's robustness in adjusting to dynamic axial motion throughout a wide range of forces. See Fig. 6(A, B, D, E) for the force tracking plots in a singular measurement and Fig. 8A for the distribution.

B. Trajectory Scanning Experiments

1) Force Tracking on UR3e Robot Arm with Admittance Controller: The UR3e robot arm without the compliant end-effector served as the baseline condition, tested at 5N with path speeds of 1cm/s and 2cm/s. The robot arm followed a zigzag path and superimposed the admittance controller in Fig. 4C as the normal velocity perpendicular to the tissue surface to maintain constant contact force. The measured mean forces, RMSE, and percentage of time above 150% of the target force for the respective path speeds can be found in Table IV. These results show that under admittance control, the robot arm struggled to maintain the target mean force but achieved lower RMSE compared to the point contact experiment. Force tracking error increased by 23.2% when path speed doubled. See Fig. 6(G, H) for the force tracking plots in a singular trial and Fig. 8B for the distribution.

2) Force Tracking on Compliant End-effector: The compliant end-effector was tested during trajectory scanning, tested at 5N with path speeds of 1cm/s and 2cm/s. The robot arm followed the path without feedback-based motion adjustments. The measured mean forces, RMSE, and percentage of time above 150% of the target force for the respective path speeds can be found in Table IV. This shows an average RMSE reduction of 68.0% compared to the robot arm only condition across both path speeds. See Fig. 6(G, H) for the force tracking plots in a singular trial and Fig. 8B for the distribution. These results show that the end-effector maintained a good performance on trajectory scanning. The RMSE at 1cm/s is comparable to the point contact experiment, with force tracking error increasing by 21.1% when path speed doubled.

TABLE IV
FORCE TRACKING PERFORMANCE COMPARISON DURING TRAJECTORY SCANNING (5N TARGET)

Path Speed	**1 cm/s**	**2 cm/s**
Robot Arm Only		
Mean (N)	6.26	6.73
RMSE (N)	2.37	2.92
% time above 150% of target	5.13	10.03
w/ Compliant End-effector		
Mean (N)	5.01	5.03
RMSE (N)	0.76	0.93
% time above 150% of target	0.00	0.00
RMSE reduction (%)	67.9	68.2

C. Supplementary Experiments

1) Force Tracking on End-effector with Porcine Tissue (Static): Force tracking on the end-effector was done on porcine tissue at target forces of 2.5N, 5N, and 10N in a stable configuration, without axial motion. The system exhibited high precision in maintaining the desired force levels, with mean values of 2.50N (RMSE: 0.03N), 4.99N (RMSE: 0.04N), and 10.00N (RMSE: 0.04N), closely matching the target forces. These results confirm the system's ability to achieve excellent force tracking with effectively no steady state error under stable conditions. Refer to Fig. 8A for the distribution.

2) Force Tracking on End-effector with Tissue Phantom (Simulated Breathing Motion): The simulated breathing conditions were tested with a tissue phantom and the same target forces. The results showed measured mean forces of 2.50N (RMSE: 0.59N), 5.02N (RMSE: 0.46N), and 10.00N (RMSE: 1.12N). These results show that the system achieved comparable performance on a homogeneous tissue phantom, suggesting that such phantoms may serve as viable alternatives to *ex vivo* tissue for preliminary testing, thereby reducing the logistical challenges of tissue procurement. Refer to Fig. 9B for the distribution.

3) Force Tracking Comparison: UR3e Robot Arm and End-effector with Porcine Tissue (Sudden Movement): Finally, force tracking was assessed in the event of a sudden 20mm sinusoidal displacement over 0.25s rise time and 0.25s fall time, both the UR3e robot arm and the compliant end-effector were tasked with maintaining a 5N contact force on porcine tissue. The end-effector showed superior recovery and less peak force, with contact forces ranging from 0.96N to 13.94N, while the UR3e arm exhibited more pronounced fluctuations, with contact forces ranging from 0.12N to 28.57N. These results show the end-effector's superior ability to maintain stable contact while not applying excessive force under abrupt motion. See Fig. 9C for the comparative tracking plot.

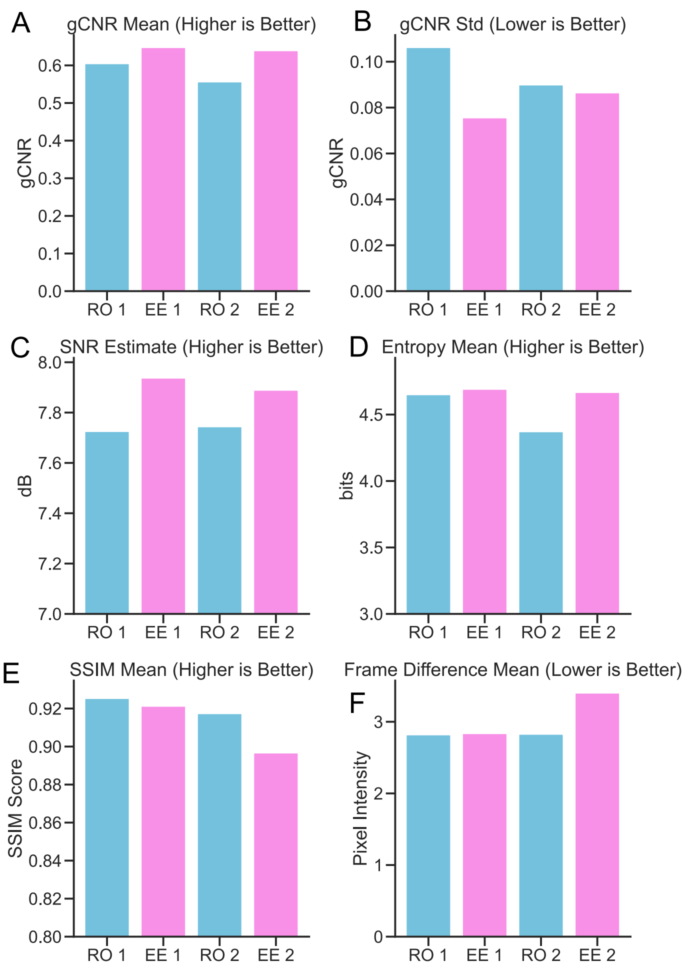

Fig. 11. Image quality and stability metrics including generalized contrast-to-noise ratio (gCNR) A: mean and B: standard deviation; C: signal-to-noise ratio (SNR); D: entropy; E: structural similarity index measure (SSIM); and F: frame difference. RO 1: robot arm only condition at 1cm/s; EE 1: compliant end-effector at 1cm/s; RO 2: robot arm only condition at 2cm/s; EE 2: compliant end-effector at 2cm/s. The compliant end-effector shows statistically superior performance in four out of six image quality metrics, indicating improved imaging quality and consistency during trajectory scanning.

D. Imaging Analysis

Although this study focused on safe force application and compliance without incorporating US imaging as control feedback, US data was collected for observational purposes to provide a more comprehensive evaluation. Despite slight differences in probe alignment between the rigid mount and compliant end-effector due to physical configuration differences, both systems successfully captured the silicone cube reference object embedded in the platform during scanning, as shown in Fig. 10. As visualized in Fig. 11, quantitative analysis was performed using six metrics: generalized contrast-to-noise ratio (gCNR) mean and standard deviation, signal-to-noise ratio (SNR), entropy, structural similarity index measure (SSIM), and frame difference. Three metrics (gCNR, SNR, and entropy) assessed image quality using mean values. SSIM and frame difference quantified stability by measuring inter-frame visual content stability, while gCNR standard deviation evaluated quality consistency. Statistical comparisons (Table V) were conducted using factorial ANOVA with Type II sum of squares to assess main effects of method (end-effector vs robot arm only) and speed (1 cm/s vs. 2 cm/s) and their interaction, with effect sizes reported as η^2, variance equality on gCNR standard deviation evaluated via Levene's test, and pairwise differences examined using independent t-tests and Mann–Whitney U tests. Overall, the results show that the end-effector outperformed the robot arm across four of the six metrics, and slower scan speeds (1 cm/s) yielded superior quality and stability compared to faster scans (2 cm/s). Qualitatively, in a blinded evaluation where comparative images from both conditions were randomly labeled A or B, a qualified clinician investigator found no significant differences in object visualization between the two acquisition methods.

TABLE V
STATISTICAL COMPARISON OF IMAGE QUALITY METRICS ACROSS METHODS AND EXPERIMENTAL CONDITIONS

	Method			Speed		
Metric	**Best**	**p-value**	η^2	**Best**	**p-value**	η^2
gCNR Mean$^\uparrow$	EE	< 0.001	0.088	1cm/s	< 0.001	0.018
gCNR Std$^\downarrow$	EE	< 0.001	0.020	2cm/s	< 0.001	0.004
SNR$^\uparrow$	EE	< 0.001	0.035	—	0.336	0.000
Entropy$^\uparrow$	EE	< 0.001	0.081	1cm/s	< 0.001	0.084
SSIM$^\uparrow$	RO	< 0.001	0.016	1cm/s	< 0.001	0.045
Frame Diff$^\downarrow$	RO	< 0.001	0.007	1cm/s	< 0.001	0.015

EE: end-effector; RO: robot arm
$^\uparrow$ Higher is better, $^\downarrow$ Lower is better
— = Not significant ($p > 0.05$)

IV. Discussion

This study developed and evaluated a QDD compliant end-effector for safe US imaging, along with a dynamic *ex vivo* testing platform. The experimental results demonstrate both contributions' effectiveness. Specifically, the end-effector showed superior force tracking accuracy during dynamic motion compared to conventional active compliance systems, achieving an average force tracking error of 0.83N RMSE across 2.5N-15N forces. This is significantly lower than the 4.70N RMSE of conventional robot arm active compliance or the 2.47N variability of manual scans [39]. Additionally, the dynamic tissue platform provided realistic surfaces and motion simulation for comprehensive evaluation.

In the image quality metrics, the compliant end-effector showed superior performance in most quality metrics compared to the robot arm. While some variability in stability metrics was observed due to the compliant mechanism's quicker adaptive behavior, this represents a design trade-off prioritizing patient safety over rigid contact. Importantly, in a blinded qualitative assessment, a qualified clinician investigator found no significant differences in the images. This finding shows that our safety-focused approach achieves quantitatively superior performance while maintaining clinically equivalent image quality. Since the main objective of this study was to enhance safety and prevent excessive force application, image quality was not incorporated as a feedback parameter in the control system. Future work could integrate image quality metrics to provide additional control feedback.

A limitation of the current system is the use of PLA plastic in constructing the end-effector, which restricted the maximum force to 15N, which is below the maximum clinical requirement of 17.3N. The PLA structure's deformation and resulting belt slippage above 15N prevent operation in the 15-20N range. While we intended the system to provide controlled slippage at high forces as a safety feature (since extrapolated clinical data indicates forces exceeding 20N could cause patient discomfort or pain [23]), the 15N mark falls short. Future iterations will address this limitation by using stiffer materials such as carbon-fiber-reinforced nylon or aluminum, allowing operation closer to the 17.3N clinical maximum while maintaining protective slippage before reaching the 20N discomfort threshold.

Future work should investigate different tissue types, movements, and scanning patterns through more comprehensive testing, ultimately progressing to in vivo patient trials. This study used simple model-free controllers to establish a baseline comparison between the end-effector and robot arm alone. Future work can implement model-informed controllers using the dynamic model from Eq. 14-16, and explore integrating the robot arm's active compliance with the end-effector's control, rather than controlling them independently as done here to isolate effects. Another direction involves implementing variable force profiles rather than maintaining constant values. Finally, although this work focused exclusively on force feedback for safety and compliance, future studies could incorporate US image feedback into the control scheme. This integration would enable real-time adjustment of scanning parameters based on image quality metrics, automatic tracking of anatomical features during respiratory motion, and optimization of probe orientation to maintain optimal acoustic windows throughout the examination.

From a clinical perspective, the end-effector's ability to maintain accurate force tracking during dynamic movement is significantly better than using conventional active compliance control on only the robot arm. The precise force tracking could lead to more consistent US images, enhancing diagnostic accuracy and reducing the need for repeat scans. Additionally, the end-effector's rapid response to sudden movements (peak force of 13.94N compared to 28.57N for the robot arm) suggests a reduced risk of patient discomfort or injury during unexpected patient movements, which is important in pediatric or geriatric care where compliance may be important.

V. Conclusion

The compliant QDD end-effector presented in this paper addresses key compliance challenges in robotic US imaging by ensuring patient safety and comfort through passive mechanical compliance while maintaining precise force control. The experiments validate its ability to achieve stable force application during simulated dynamic conditions that mimic respiratory motion, which is essential for abdominal and retroperitoneal imaging.

Compared to existing compliant end-effectors, this design offers distinct advantages: it provides position feedback and independent force regulation unlike soft fluidic systems [17], achieves a wider force range with superior static regulation compared to pneumatic designs [8], and accommodates a substantially larger operating position range than linear elastic actuators [18], [19]. This is the first compliant end-effector, to our knowledge, validated on a moving tissue platform for robotic US, making direct dynamic performance comparisons with other systems challenging.

This strengths of the proposed system make it well-suited for clinical applications and is a significant step toward safer, more reliable robotic US systems.

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